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(54) **Titre : FORMES CRISTALLINES DE TROFINETIDE**
(54) **Title: CRYSTALLINE FORMS OF TROFINETIDE**

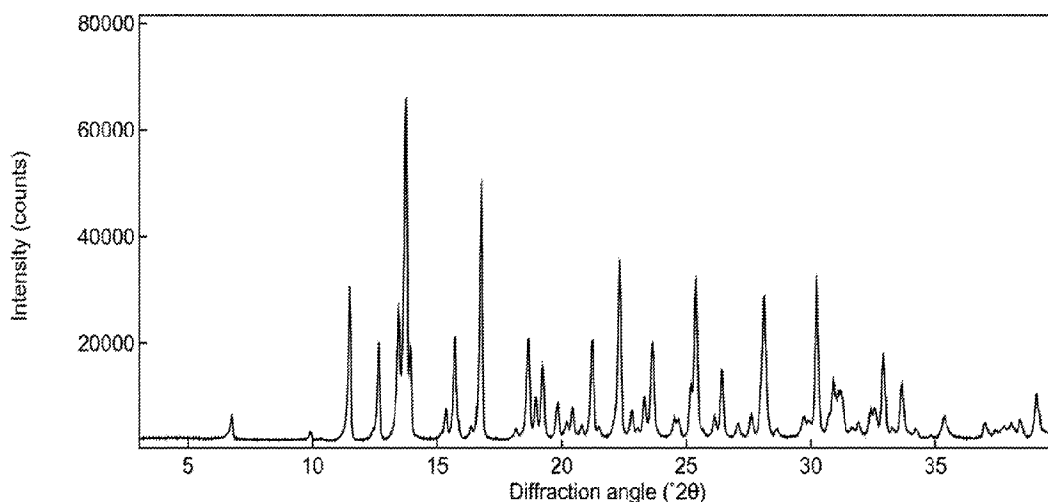


Fig. 1

(57) **Abrégé/Abstract:**

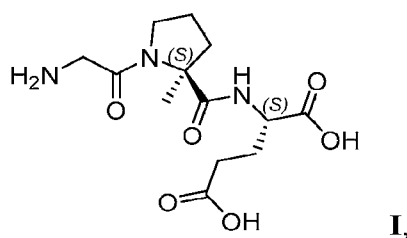
This disclosure provides crystalline forms of trofinetide and trofinetide hydrate, pharmaceutical compositions comprising crystalline forms of trofinetide and trofinetide hydrate, methods of making crystalline forms of trofinetide or trofinetide hydrate, and methods of treating a disease, condition, or disorder in a subject comprising administering a composition comprising crystalline forms of trofinetide or trofinetide hydrate to the subject.



- [0106] In another embodiment, the method of making Form A further comprises isolating, e.g., by filtration, the solid thus obtained to give a wet cake comprising Form A.
- [0107] In another embodiment, the method of making Form A further comprises washing the wet cake comprising Form A with precooled ethanol at about 0 °C.
- [0108] In another embodiment, the method of making Form A further comprises drying the wet cake comprising Form A under vacuum.

VII. Definitions

- [0109] The term "trofinetide" as used herein refers to glycyl-L-2-methylprolyl-L-glutamic acid of Formula I:



- wherein each stereocenter is in the *S* configuration. The IUPAC name of trofinetide is (2*S*)-2-[[[(2*S*)-1-(2-aminoacetyl)-2-methylpyrrolidine-2-carbonyl]amino]pentanedioic acid. Trofinetide is also referred to as "G-2-MePE," "H-Gly-MePro-Glu-OH," or "Gly-MePro-Glu-OH."
- [0110] Trofinetide may form crystalline solids that incorporate solvates, e.g., water or methanol, into the crystal lattice without chemical alteration of the trofinetide molecule. The term "trofinetide hydrate" is represented by the formula: trofinetide · xH₂O, wherein x is the ratio of H₂O moles per mole of trofinetide. Trofinetide hydrate does not have to contain water in stoichiometric amounts, e.g., x can be about 2.5. In one embodiment, x is about 1 to about 5. In another embodiment, x is about 2 to about 4. In another embodiment, x is about 2.5 to about 3.5. In another embodiment, x is about 2. In another embodiment x is about 2.5. In another embodiment x is about 3. In another embodiment, x is about 3.5. In another embodiment, x is about 4. In another embodiment, x is about 4.5.
- [0111] As used herein, the term "substantially pure" with reference to a trofinetide polymorph means that the crystalline material comprises about 10% or less, e.g., about 1% to about 10%, e.g., about 9%, about 8%, about 7%, about 6%, about 5%, about 4%, about 3%, about 2%, or about 1%, by weight of any other crystalline or amorphous form(s) of

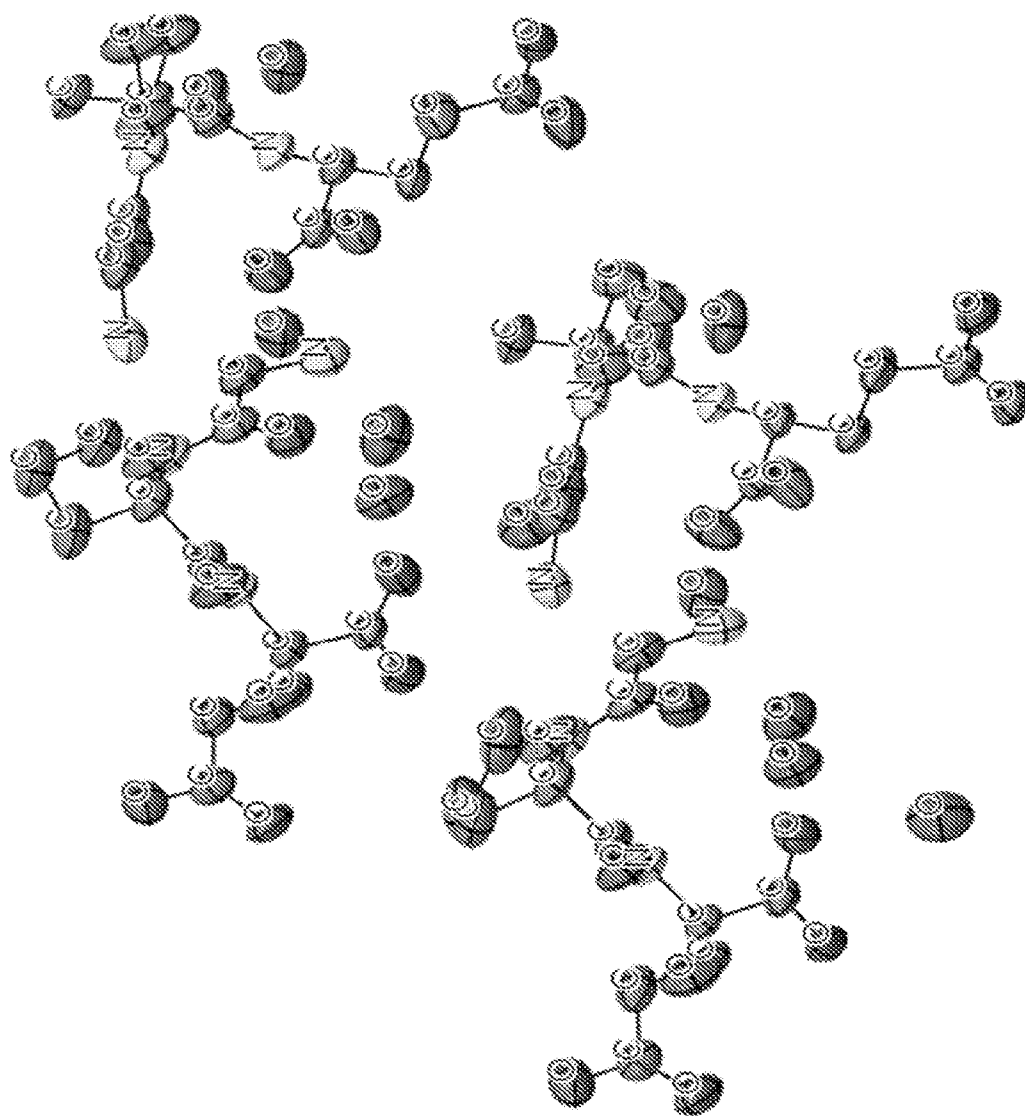


Fig. 8

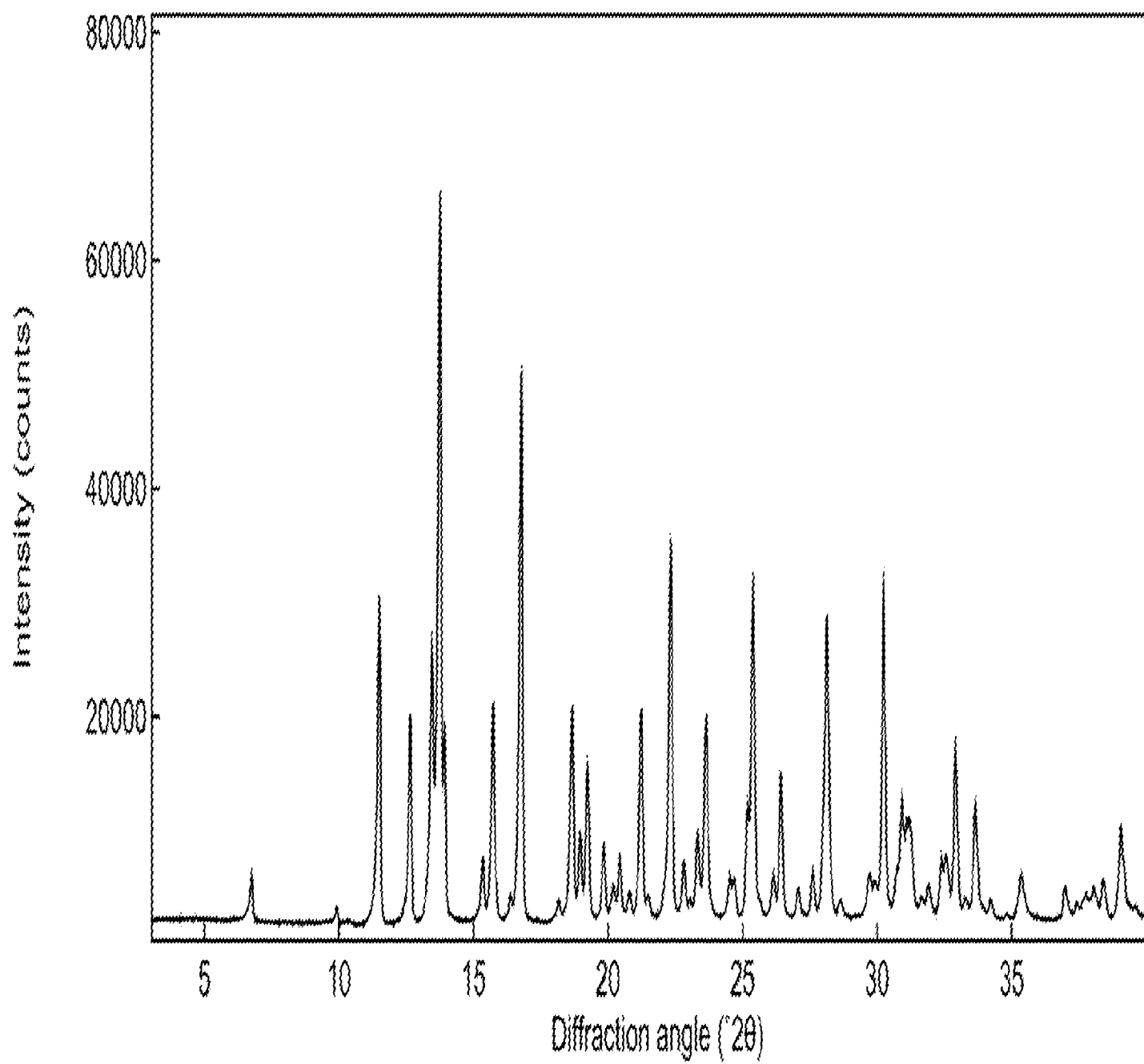


Fig. 1